Afternoon Lab

Build and Evaluate Your Own RAG System

Three-hour practical extension for students who have completed the guided notebook

Work individually or in small groups. Make a copy of the guided notebook before starting.

Learning goals

- Adapt an existing RAG pipeline to a new document set.
- Test whether retrieval finds the required evidence.
- Evaluate answerability, citation behaviour and end-to-end performance.
- Investigate at least one failure and explain what caused it.
- Optionally use an LLM-as-a-judge to assess semantic answer quality.

Recommended data

Choose a small, factual and text-based document set so that the focus remains on RAG rather than data cleaning.

- Use 2-5 short PDFs.
- Prefer text-based PDFs rather than scanned images.
- Choose documents with clear facts, rules, limits or procedures.
- Do not use confidential, personal or sensitive information.
- Suitable examples include public policies, module handbooks, product manuals, public reports and game rules.

Task ladder

Complete the tasks in order. The first five tasks form the core lab. Tasks 6 and 7 are extensions.

Task 1 - Change the knowledge base

- Replace the original PDFs with a small new document set.
- Rebuild the vector store.
- Run the ingestion checks.
- Ask at least three factual questions.

Goal: understand which parts of the notebook are reusable.

Task 2 - Create an unanswerable question

- Add a question whose answer is not present in the documents.
- Check whether the model sets answerable to False.
- Check that the answer does not contain unsupported citations.

Goal: understand abstention and hallucination risk.

Task 3 - Test retrieval settings

- Run the same question with different values of top_k, such as 1, 3 and 5.
- Record the retrieved sources and first relevant rank.
- Observe whether the final answer changes.

Goal: see how retrieval choices affect generation.

Task 4 - Test chunking choices

- Rebuild the store using two chunk configurations.
- For example, compare chunk_size=500, chunk_overlap=50 with chunk_size=1000, chunk_overlap=200.
- Run the same questions against both stores and compare the results.

Goal: learn that chunking is a design choice rather than a fixed rule.

Task 5 - Build a mini evaluation set

- Create 3-5 golden evaluation cases.
- Include a direct factual question, a paraphrased question and an unanswerable question.
- Where possible, add a boundary or exception case.
- Run retrieval and end-to-end evaluation for each case.

Goal: practise systematic evaluation rather than random testing.

Task 6 - Investigate a failure

- Find or create one case where the system performs poorly.
- Identify whether the failure came from ingestion, chunking, retrieval, prompting, generation or citation handling.
- Make one change and rerun the case.

Goal: develop practical debugging skills.

Task 7 - Optional: LLM-as-a-judge

- Run the judge on one clearly correct answer.
- Run it again on a deliberately weakened or incorrect answer.
- Compare the score and explanation with your own assessment.
- Remember that the judge score is a model-based assessment, not objective ground truth.

Goal: understand what an LLM judge can and cannot measure.

Suggested schedule

Time	Activity
0-30 min	Choose documents, update paths and complete ingestion.
30-75 min	Retrieve evidence and test factual and unanswerable questions.
75-120 min	Create the mini evaluation set and run metrics.
120-150 min	Investigate one failure or try the LLM judge.
150-180 min	Tidy the notebook and write the reflection.

Minimum success criteria

- The vector store contains the expected number of chunks.
- At least three evaluation cases are completed.
- At least one case is intentionally unanswerable.
- Retrieval metrics are calculated for the evaluation cases.
- Citation validation runs successfully.
- The notebook includes one successful case and one failed or difficult case.

Deliverables

- The completed notebook.
- The documents used for the knowledge base.
- The evaluation cases.
- A short reflection of approximately 150-250 words.

Reflection prompts

1. Which questions retrieved the correct evidence?
2. Which questions failed, and what caused the failure?
3. Did changing top_k or chunk size affect the results?
4. Did the system abstain correctly when evidence was missing?
5. Did the LLM judge agree with your own assessment?
6. What would you improve if you had more time?

Teaching assistants are available to help with environment problems, document loading, debugging and interpretation of evaluation results.